SFSANet: Multiscale Object Detection in Remote Sensing Image Based on Semantic Fusion and Scale Adaptability

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***Abstract*— In the field of computer vision, remote sensing image object detection plays an important role. Although the object detection algorithm has made significant progress, there are still problems in detecting objects with multiscale in remote sensing image. Due to the insufficient utilization of object feature information, the detection accuracy of multiscale objects is very low. To address the aforementioned issues, this article proposes an effective object detection algorithm for remote sensing image based on semantic fusion and scale adaptability (SFSANet). First, in view of the problem that the existing methods ignore the semantic differences between different scale feature maps, the semantic fusion (SF) module is proposed to enrich the semantic information and improve the ability to classify and locate objects. Next, to address the issue of the objects being easily interfered in complex background and the detection performance is poor, the spatial location attention (SLA) module is constructed to suppress background information and make key objects more prominent. Additionally, the scale adaptability (SA) module is designed to enrich the expression of feature information, realize the integration of global and local information, and ensure the integrity of image structure. Finally, we adopt the SIoU loss function as the localization loss to expedite model convergence. In order to verify the effectiveness of the proposed method, we conduct experiments on the mainstream datasets DIOR and NWPU VHR-10, which fully demonstrate the superiority of the proposed method.**

***Index Terms*— Object detection, receptive field, remote sensing image, semantic fusion (SF).**

I. Introduction

AS HIGH-RESOLUTION remote sensing images can be easily obtained, accurately identifying and locating objects of interest are crucial for aerial and satellite image processing [1], [2], [3], [4]. Remote sensing image object detection has a wide range of application values [5], [6], [7], such as crop detection, environmental protection, urban planning, public safety, and military investigation. This topic has been studied in the field of earth observation for several decades and has gradually become an important research topic in the field of computer vision.



Fig. 1. Examples of remote sensing image. To generate clear visuals, we have hidden label information and employed rectangular boxes of different colors to mark various object categories. It can be seen from the examples that the same type of objects show obvious scale differences.

The flexible flight altitude of the remote sensing platform results in rapid changes in the distance between the platform and the shooting object. Coupled with the variable shooting angle, remote sensing images often include both near and far objects [8], [9], [10], [11], [12]. This allows for a variety of scales in the object, as shown in Fig. 1. Therefore, the object detection algorithms in natural scenes do not perform well when directly applied to remote sensing images. Thus, we need to develop a detection algorithm specifically for remote sensing images to solve the challenges brought by the difference in object scale.

In general, existing object detection methods can be divided into two categories: traditional algorithms and algorithms based on deep learning. The traditional object detection method is based on shape and texture information [13], salient features [14], and scale-invariant features [15], using handcrafted features to represent and locate specific objects. However, these traditional algorithms often suffer significant interference from complex backgrounds, resulting in poor detection performance.

Compared with traditional algorithms, the object detection algorithm based on deep learning can extract deep feature representations rich in semantic information to have excellent classification and localization performance. At present, the mainstream object detection algorithms based on deep learning can be divided into two categories: anchor-based

Manuscript received 12 February 2024; revised 21 March 2024; accepted 8 April 2024. Date of publication 11 April 2024; date of current version 22 April 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 61702347 and Grant 62027801, in part by the Natural Science Foundation of Hebei Province under Grant F2022210007 and Grant F2017210161, in part by the Science and Technology Project of Hebei Education Department under Grant ZD2022100 and Grant QN2017132, in part by the Central Guidance on Local Science and Technology Development Fund under Grant 226Z0501G, and in part by the Graduate Innovation Program under Grant YC202450. *(Corresponding author: Yunzuo Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3387572

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algorithms and anchor-free algorithms [16], [17], [18], [19]. The anchor-based detection algorithms can be further divided into two-stage algorithms [20], [21], [22], [23], [24], [25] and single-stage algorithms [26], [27], [28], [29], [30], [31]. While the two-stage algorithms offer high detection accuracy, their inference speed is relatively slow, which greatly hinders their practical applications in real-life scenarios. Therefore, we choose the single-stage detection algorithm as the baseline model to realize multiscale object detection in remote sensing images.

To address the issue, many researchers have focused on feature fusion [32], [33]. Chen et al. [34] designed a parallel dual-fusion feature pyramid structure, which includes a bottom-up fusion module capable of simultaneously detecting small and large objects while achieving higher accuracy. Wang et al. [35] focused on the scale issue and proposed a special module to fuse context in multiscale features. Xu et al. [36] attributed the poor detection performance of objects with large aspect ratios and different scales to the feature misalignment problem and proposed a feature alignment detection method. Yi et al. [37] proposed an efficient and lightweight multiscale feature fusion network, which achieved a better balance between counting accuracy and running speed. Lu et al. [38] proposed an end-to-end network based on attention and feature fusion for SSD. While the above-mentioned methods have made some progress, they ignore the differences in semantic information between different scale feature maps, which are crucial for object detection [39], [40], [41]. Furthermore, the presence of complex backgrounds can introduce interference into the process of object detection. To solve the above problems, we propose multiscale object detection in remote sensing images based on semantic fusion (SF) and scale adaptability (SA). We conduct extensive experiments in mainstream datasets to verify the effectiveness of the proposed method. Our main contributions are as follows.

1) The SF module is proposed to reduce the semantic difference between different scale feature maps. This module calculates the correlation between pixels within the same feature map and between pixels in different scale feature maps through matrix multiplication operations.
2) The spatial location attention (SLA) module is constructed to suppress background information and highlight key objects. This module integrates spatial information and dimensional information, which can not only improve the spatial structure perception of the model but also increase the sensitivity to the object.
3) The SA module is designed. Through the reuse of features, the output of each branch is connected to the input of the next branch, which enhances the extraction and localization of detailed features and enriches the variety of receptive fields.
4) Experimental results on mainstream datasets show that the proposed method exhibits superior advantages compared with other detection methods.

The remaining structure of this article is as follows. Section II discusses related techniques. In Section III, the proposed approach is described in detail. Section IV conducts extensive experiments to verify the effectiveness of the proposed method. Finally, the conclusion is provided in Section V.

II. Related Work

To address the above-mentioned challenges, we will focus on introducing three important techniques in this section: semantic information exploitation, visual attention mechanism, and the receptive field enhancement strategies.

A. Semantic Information Exploitation

Semantic information is widely used for a wide variety of tasks such as object detection [5], [6], [7], image segmentation [42], and video segmentation [43]. Many researchers are dedicated to exploring different methods to leverage semantic information for enhancing the performance of object detection. Semantic information helps to explain the objects and backgrounds contained in the image and the specific relationships between them. Tian et al. [39] and Luo et al. [40] also demonstrate the importance of using semantic information for object detection. Therefore, extracting semantic information for remote sensing images with complex backgrounds is crucial.

For example, Lin et al. [32] proposed using the feature pyramid network (FPN) for object detection. FPN uses both the high resolution of the shallow feature map and the semantic information of the deep feature map and uses the top-down path to fuse the semantic information of the deep feature map with the spatial information of the shallow feature, so as to improve the accuracy of object detection. Dai et al. [33] devised a feature fusion method that combines bottom-up and top-down approaches to enhance the localization and semantic features of multiscale ships, particularly small-sized ships. Liang et al. [44] proposed a spatial context analysis method to solve the problem of small objects in UAV images, which is effective for multiclass small object detection. Wen et al. [45] leveraged the spatiotemporal context to design adjacent context loss to penalize the large displacement of the object in the time domain for training the network. However, the existing object detection algorithms ignore the semantic differences between feature maps at different scales. To solve the above problems, we design the SF module to reduce the semantic difference between different scale feature maps and enrich the expression of semantic information.

B. Visual Attention Mechanism

The visual attention mechanism is the neural mechanism in the human brain used to selectively focus on and process specific visual information while ignoring other irrelevant information. In deep learning, visual attention is introduced into neural networks to simulate the human visual attention process, thereby improving the robustness of the network. At present, the mainstream attention mechanism can be divided into channel attention mechanism, spatial attention mechanism, and fusion attention mechanism [46].

Channel attention can help the model better focus on interactions and important differences between different channels.

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Among them, the squeeze-and-excitation network (SENet) [47] is the most popular channel attention mechanism. However, SENet only considers the encoded information between channels and ignores the importance of location information. Spatial attention can help the model pay attention to information in different locations and improve their perception of spatial structures. Spatial transformer network (STN) [48] is the most representative example. However, STN lacks information interaction between channels and only focuses on information at different locations. Therefore, researchers combine channel attention and spatial attention to improve the expression and perception of features. Among them, the convolutional block attention module (CBAM) [49] is the most classic representative. CBAM reduces the channel dimensions and employs convolution to compute spatial attention for integrating positional information. However, convolution can only capture local relationships, making it difficult to model global information. Considering that remote sensing images contain numerous objects and have complex backgrounds, it is necessary to model global information and promote interaction between global and local positions [50]. Therefore, we construct the SLA module to facilitate mutual influence among global spatial positions, fully leveraging the utilization of contextual information.

C. Receptive Field Enhancement Strategies

In object detection, the receptive field refers to the area of the neural network that neurons at a specific location "see" on the input image [51]. The size of the receptive field significantly affects whether the neural network can effectively capture objects of different sizes. Larger receptive fields enable feature maps to cover more global information, while smaller receptive fields contain less information [52]. Therefore, the receptive field is one of the important factors affecting object detection. To enhance the receptive field, various approaches can be considered. One approach is to add convolution layers or use a larger-sized convolutional kernel. However, as the depth of the network deepens, the problem of gradient disappearance may occur, and the number of parameters in the model will increase. Another approach is to enhance the receptive field by adding the pooling layer, but this tends to lose useful information in the feature.

To address the aforementioned issues, the Google team [53] devised a sparse inception network structure that successfully reduced the number of parameters while enhancing the performance of the neural network. Inspired by the receptive field structure in the human visual system, Liu et al. [54] proposed a novel receptive field module, which added dilated convolution layer based on inception to effectively increase the receptive field without significantly increasing the number of parameters. Li et al. [55] introduced a dynamic selection mechanism for convolutional neural networks, allowing each neuron to adaptively adjust the size of its receptive field based on input information at multiple scales. Jang et al. [56] presented a learnable dynamically optimized pooling operation that adjusts size. This operation identifies the optimal scale factor and receptive field for intermediate feature maps and propagates this information to subsequent layers, achieving scale optimization across the entire network. However, these methods have certain limitations, as they cannot guarantee the integrity of image structures and may not fully leverage the features of the objects. Therefore, to enhance the perception of objects at different scales and achieve integration of global and local information, we design the SA module. Through this module, we aim to comprehensively capture the features of objects, ultimately improving performance in object detection tasks.

III. Proposed Method

In this section, four key components will be detailed: the SF module, the SLA module, the SA module, and the loss function. Besides, the overall framework of the proposed method is shown in Fig. 2, which consists of three parts: the backbone, the neck, and the detection head.

A. Semantic Fusion

Existing object detection methods usually use feature pyramids and their variants to fuse feature maps of different scales, but they often ignore the semantic differences between feature maps of different scales. In the FPN, shallow feature maps have high resolution and contain more location and detail information. On the contrary, deep feature maps have stronger semantic information, lower resolution, and poor perception of detail. Given the semantic differences between feature maps of different scales, the SF module is designed.

As shown in Fig. 3, the module uses matrix multiplication operation to calculate the correlation between pixels in the same feature map and the correlation between pixels in different scale feature maps to improve the detection performance of the object. For the shallow feature map with high resolution, first, two new feature maps $A, B \in \mathbb{R}^{C\times H\times W}$ are generated through the convolution layer. Next, the feature map A is reshaped and transposed to obtain the feature map $M \in \mathbb{R}^{N\times C}$, while the feature map B is reshaped to obtain the feature map $V \in \mathbb{R}^{C\times N}$. Furthermore, M and V are dotted so that each pixel in the feature map is associated. Finally, the softmax activation function is applied to the nonlinear mapping of the result after dot multiplication to obtain S_{ij}. The higher the similarity between the two locations, the larger the mapping result value. The mapping results are calculated as follows:

$$S_{ij} = \frac{\exp(M_j \cdot V_i)}{\sum_{j=1}^{N} \exp(M_j \cdot V_i)} \tag{1}$$

where S_{ij} represents the influence of position j on position i. N represents the number of pixels. For the deep feature map with low resolution, a new feature map $C \in \mathbb{R}^{C\times H\times W}$ is also generated through the convolution layer. Then, the feature map C is reshaped and the dot multiplied with the feature map S_{ij}, and the result of dot multiplication is reshaped to obtain the feature map $F \in \mathbb{R}^{C\times H\times W}$. Finally, the elementwise sum of F and C yields the output O

$$O_i = \gamma \sum_{j=1}^{N} (S_{ij} v_j) + C_i \tag{2}$$

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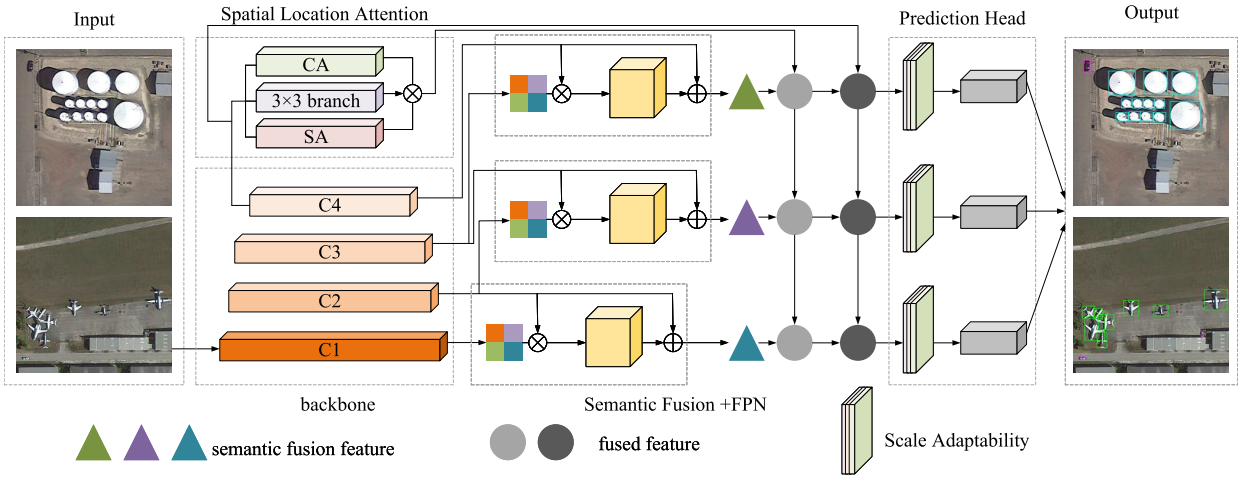


Fig. 2. Overall framework of the proposed method SFSANet. Remote sensing images are fed into the backbone network. Then, the extracted feature maps are fed into the designed SLA module. Next, the neck realizes multiscale feature fusion through the proposed SF and feature pyramid. Finally, the prediction head extracts multiscale features through the proposed SA module to predict the object category and bounding box.

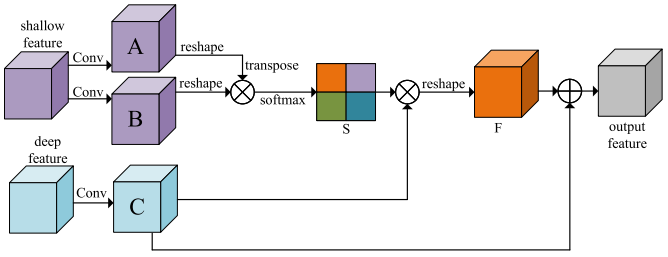


Fig. 3. SF structure diagram. This module realizes the smooth fusion of deep and shallow feature maps and effectively overcomes the semantic differences between feature maps of different scales.

where γ is the learnable parameter, initialized to 0. v_j represents the feature mapping of the jth position of the shallow feature, and C_i represents the feature mapping of the ith position of the deep feature. According to the formula, we use matrix multiplication to measure the correlation of pixels in the feature map. The resulting feature map is derived from the deep feature map, so it retains rich semantic information, which is helpful for object detection and location.

B. Spatial Location Attention

In the object detection task of remote sensing image, the fusion of spatial information and dimensional information can produce more discriminative features, so that the model can focus on the important information. Coordinate attention (CA) embeds positional information into channel attention and captures long-range information in both horizontal and vertical directions. However, CA neglects the interaction of global spatial locations and hinders the local cross-channel interaction and the utilization of context information. Inspired by CA, spatial location attention is constructed to better capture key features in image.

The SLA module is a parallel structure, consisting of CA, 3×3 branch, and spatial attention, as shown in Fig. 4. For the CA part, CA performs global average pooling of input feature maps according to width and height, to obtain feature maps along the width and height directions, respectively. The output of the feature maps of the two directions in the channel c is as follows:

$$z_c^h(h) = \frac{1}{W} \sum_{0 \le j \le W} x_c(h, i) \tag{3}$$

$$z_c^w(w) = \frac{1}{H} \sum_{0 \le j \le H} x_c(j, w) \tag{4}$$

where z_c^h and z_c^w represent the output height and output width of channel c, respectively. $x_c(h, i)$ denotes the value of the feature map at height h and width coordinate i for the cth channel; $x_c(j, w)$ denotes the value of the feature map at width w and height coordinate j for the cth channel. W and H represent the width and height of the feature map, respectively. After that, the direction-aware feature maps generated from the pooling are concatenated and fed into a 1×1 convolution to reduce the dimension, and the swish activation function is used to obtain the feature map $f \in R^{C/r \times (H+W)}$

$$f = \text{Swish}(F([Z^h, Z^w])) \tag{5}$$

where $[\cdot, \cdot]$ denotes concatenation along the spatial dimension. Then, the feature map f is passed through the 1×1 convolution to obtain feature maps F_w and F_h, and activation functions are applied to obtain the weights for the width and height directions

$$g^w = \text{sigmoid}(F_w(f^w)) \tag{6}$$

$$g^h = \text{sigmoid}(F_h(f^h)) \tag{7}$$

where g^w and g^h represent weights in the direction of width and height, respectively. Finally, the output is obtained by weighting the input feature map and the feature map in both directions

$$y_c(i, j) = x_c(i, j) \times g_c^w(i) \times g_c^h(j). \tag{8}$$

For the 3×3 branch, only a single 3×3 convolution layer is stacked, and normalization and activation functions are added to improve the stability of the model and make it

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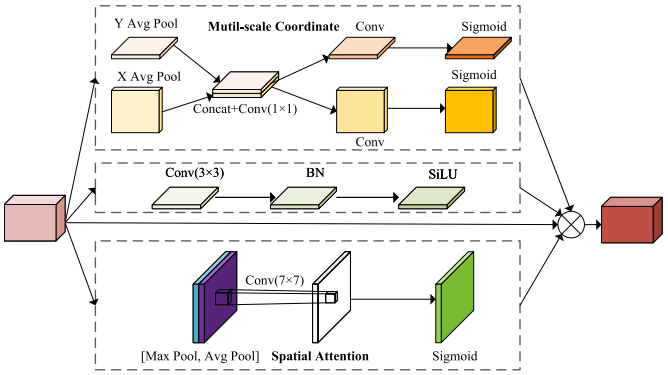


Fig. 4. SLA structure diagram. This module uses a parallel structure to build cross-channel and cross-space information interaction and improve the efficiency of feature extraction and the sensitivity of the model to the object.

better adapt to complex contextual information. Its calculation formula is as follows:

$$z = \mathrm{SiLU}(\mathrm{BN}(\mathrm{Conv}^{3\times3}(x_c))). \tag{9}$$

For the spatial attention part, the correlation of global spatial locations is established to make the model pay more attention to the object region. Specifically, two $1 \times H \times W$ feature maps are obtained by global maximum pooling and global average pooling of input features, which are stacked into $2 \times H \times W$ feature maps. Next, a 7×7 convolution is applied to obtain the $1 \times H \times W$ feature map. Finally, the feature map of spatial attention is obtained through the activation function

$$m_s = \mathrm{sigma}\big(\mathrm{Conv}^{7\times7}\big(\big[\mathrm{Avg}(x_i), \mathrm{Max}(x_i)\big]\big)\big) \tag{10}$$

where σ represents the sigmoid activation function and $[\cdot, \cdot]$ indicates the concatenation operation along the channel dimension. Thus, the attention weight of the spatial coordinate is

$$w_s = y_c \times z \times m_s. \tag{11}$$

C. Scale Adaptability

In the task of object detection, local information and global information play an important role in object detection. In order to more effectively capture information about the object and enhance the utilization of object feature information, the SA module is constructed. By adding SA to the output branch, the accuracy of multiscale prediction is improved. SA consists of standard convolution and dilated convolution, and this combination helps retain crucial critical details while also capturing global information. Among them, dilated convolution influences the sampling positions and density by introducing fixed-size interval into the convolution kernel. By adjusting the dilation factor, the range of the receptive field can be altered, so that the module can be more adapted to the remote sensing image scene. The dilated convolution kernel is calculated as follows:

$$k_d = k_s + (k_s - 1)(r - 1) \tag{12}$$

where k_d represents the size of the dilated convolution kernel, k_s represents the size of the standard convolution kernel, and r is the dilation rate. The formula for calculating receptive field after dilated convolution is as follows:

$$r_n = r_{n-1} + (k_d - 1)\prod_{i=1}^{k-1} s_i \tag{13}$$

where r_n represents the receptive field of the nth layer, r_{n-1} represents the receptive field of the $n-1$ layer, and s_i represents the stride of the ith layer. The size of the output feature map after dilated convolution is

$$O = \left\lfloor \frac{I + 2\mathrm{p} - k_d}{\mathrm{s}} \right\rfloor + 1 \tag{14}$$

where O represents the size of the output feature map, I represents the size of the input feature map, p represents the padding size for the input feature map, s represents the stride of the convolution kernel, and $\lfloor . \rfloor$ represents the floor operation.

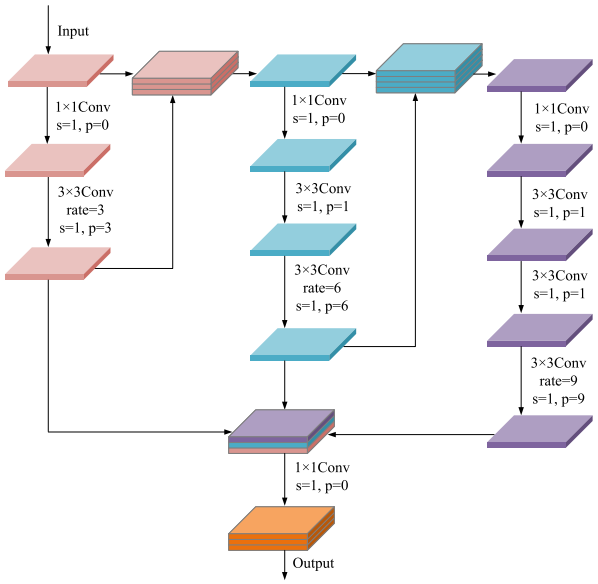


Fig. 5. SA structure diagram. This module enriches the types of receptive fields and strengthens the perception of multiscale objects.

As shown in Fig. 5, this module consists of three branches. First, the input and output of each branch are concatenated to become the input for the next branch. Second, in the input of each branch, the 1×1 convolution is used to reduce the number of channels, so as to reduce the number of parameter in the model. In the leftmost branch, a 3×3 dilated convolution layer is designed. In the middle branch, a 3×3 convolutional layer and a 3×3 dilated convolutional layer are combined. In the rightmost branch, it contains two 3×3 convolutional layers and one 3×3 dilated convolutional layer. Finally, the output of the three branches are concatenated, and the number of channels is adjusted by the 1×1 convolution. The construction of this module further expands the receptive field, enhances detailed feature extraction and localization, and enriches the acquisition of global information at the same time.

D. Loss Function

The role of the loss function is to guide the model during the training process to learn how to correctly classify

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detected objects, accurately locate the location of the object, and estimate the confidence in the presence of objects. This ensures that the model can effectively detect and locate objects in images during testing. The loss function of the baseline network is shown in the following equation:

$$L = \alpha L_{\text{Obj}} + \beta L_{\text{Cls}} + \gamma L_{\text{Box}} \tag{15}$$

where L represents the total loss and L_{Obj}, L_{Cls}, and L_{Box} represent confidence loss, classification loss, and location loss, respectively. To balance the loss, weights are assigned to each of them in front of the respective losses, in the following order α, β, and γ. In the baseline model, the CIoU loss function is used to calculate the localization loss. However, it uses relative values to describe the aspect ratios of the ground truth boxes and predicted boxes, which may not effectively reflect the actual differences in width and height. Therefore, we employ the SIoU loss function to calculate the localization loss, as shown in the following equation:

$$L_{\text{SIoU}} = 1 - \text{IoU} + \frac{\Delta + \Omega}{2} \tag{16}$$

where IoU represents the intersection over union between the predicted box and the ground truth box, Δ is the distance loss, and Ω is the shape loss. The cross-entropy loss function is used to calculate confidence loss and classification loss. It is defined as shown in the following equation:

$$L_{\text{BEC}} = -L \log P - (1 - L) \log(1 - P) \tag{17}$$

where L represents label confidence and P represents prediction confidence.

IV. Experiments and Results

In this section, we first introduce two popular public datasets. Then, the evaluation metrics and implementation details used in the experiment are presented. Finally, ablation experiments and comparative experiments are analyzed in detail.

A. Datasets

1) DIOR Dataset [57]: DIOR is the largest optical remote sensing object detection dataset, which includes 23 463 optical images and 192 472 object instances. The dataset contains 20 object categories, which are represented by c1–c20 in the experiment: airplane (c1), airport (c2), baseball field (c3), basketball court (c4), bridge (c5), chimney (c6), dam (c7), expressway service area (c8), expressway toll station (c9), golf course (c10), ground track field (c11), harbor (c12), overpass (c13), ship (c14), stadium (c15), storage tank (c16), tennis court (c17), train station (c18), vehicle (c19), and windmill (c20).

2) NWPU VHR-10 Dataset [58]: The NWPU VHR-10 dataset includes 800 high-resolution images selected from Google Earth and the Vaihingen databases, of which 650 images contain objects and the remaining 150 images do not contain objects. And experts manually annotated ten common object categories, which include airplane (AP), ship (SH), storage tank (ST), baseball diamond (BD), tennis courts (TC), basketball court (BC), ground track field (GTF), harbor (HA), bridge (BR), and vehicle (VE).

B. Evaluation Metrics

In the experiments, we use widely used evaluation metrics, including average precision (AP) and mean AP (mAP), to assess the accuracy of the model. Precision reflects the proportion of prediction positives correctly identified in all predicted positives. Recall indicates the proportion of TPs among all the actual positives. P and R are calculated as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{19}$$

AP and mAP are defined as follows:

$$\text{AP} = \int_0^1 P(R) dR \tag{20}$$

$$\text{mAP} = \frac{1}{N_{\text{cls}}} \sum_{i=1}^{N_{\text{cls}}} \text{AP}_i \tag{21}$$

where N_{cls} represents the total number of categories.

C. Implementation Details

In this article, all experiments are based on the PyTorch framework. The experimental runtime environment used is as follows: the NVIDIA Tesla V100S-PCIE-32GB GPU, the Intel[1] Xeon[1] Gold 6226R CPU @ 2.90 GHz, and the CentOS operating system. The initial learning rate, momentum, weight decay, and batch size are set to 0.01, 0.937, 0.0005, and 4, respectively. In the training process, the resolution of the input image of the DIOR dataset is 800 × 800. On the NWPU VHR-10 dataset, the image resolution is 1024 × 1024. In the experimental process, the DIOR dataset is divided into train, validation, and test, each containing 5862, 5863, and 11 738 images, respectively. The NWPU VHR-10 dataset only uses 650 images containing objects, with 80% of these images randomly selected for the train set, and the remaining 20% for the test set.

D. Comparison Analysis

To verify the effectiveness and robustness of the proposed algorithm SF and scale adaptability (SFSANet), we compare with state-of-the-art methods on DIOR and NWPU VHR-10 datasets.

1) Detection Results on DIOR: The performance of the proposed method on the DIOR test set compared with state-of-the-art methods is shown in Table I. The data show that our method achieved significant performance, reaching 76.6% mAP, which is notably superior to other methods. Furthermore, the proposed method shows significant improvements in certain categories, such as baseball field and bridge. Specifically, for baseball fields, stadiums, storage tanks, and vehicles, SFSANet performs well in mAP, reaching 87.8%, 83.4%, 79.4%, and 67.9%, respectively. Compared to the second-ranked model (with mAP values of 82.6%, 80.9%, 68.4%, and 55.4%), SFSANet exhibits improvements of 5.2%,

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TABLE I
COMPARISON RESULTS ON DIOR DATASET. THE BEST RESULT IN EACH CATEGORY IS HIGHLIGHTED IN BOLD

	c1/c11	c2/c12	c3/c13	c4/c14	c5/c15	c6/c16	c7/c17	c8/c18	c9/c19	c10/c20	mAP
Faster R-CNN[24]	53.6/56.9	49.3/50.2	78.8/50.1	66.2/27.7	28.0/73.0	70.9/39.8	62.3/75.2	69.0/38.6	55.2/23.6	68.0/45.4	54.1
YOLOv3[26]	72.2/61.1	29.2/44.9	74.0/49.7	78.6/87.4	31.2/70.6	69.7/68.7	26.9/87.3	48.6/29.4	54.4/48.3	31.1/78.7	57.1
CornerNet[18]	58.8/79.5	84.2/26.1	72.0/60.6	80.8/37.6	46.4/70.7	75.3/45.2	64.3/84.0	81.6/57.1	76.3/43.0	79.5/75.9	64.9
CF2PN[59]	78.3/77.1	78.3/56.8	76.5/58.7	88.4/76.1	37.0/70.6	71.0/55.5	59.9/88.8	71.2/50.8	51.2/36.9	75.6/86.4	67.3
CSFF[60]	57.2/81.6	79.6/50.7	70.1/59.5	87.4/73.3	46.1/63.4	76.6/58.5	62.7/85.9	82.6/61.9	73.2/42.9	78.2/86.9	68.0
YOLOv5s	78.3/73.1	65.0/58.3	74.3/57.4	**90.6/91.8**	44.3/67.9	80.1/82.7	48.9/89.1	57.7/49.7	63.2/55.4	68.6/78.1	68.7
SRAFNet[61]	**88.4**/82.8	76.5/16.2	**92.6**/58.0	87.9/59.4	35.8/80.9	**83.8**/55.6	58.6/**90.6**	86.8/52.0	66.8/53.2	76.4/**91.0**	69.7
SB-MSN[62]	79.6/82.4	82.2/51.8	76.4/60.8	89.8/74.4	45.6/79.7	78.2/66.4	64.8/85.6	58.9/65.4	59.3/45.1	79.2/79.9	70.3
GLNet[7]	62.9/83.0	83.2/51.8	72.0/62.6	81.1/72.0	**50.5**/75.3	79.3/53.7	67.4/81.3	86.2/65.5	70.9/43.4	81.8/89.2	70.7
ASDD[36]	85.6/78.6	82.4/62.0	75.8/58.0	89.5/84.9	40.7/76.7	77.6/65.3	64.7/87.9	67.1/62.4	61.7/44.5	80.8/76.3	71.1
YOLOv7[28]	85.0/78.2	79.0/58.9	80.1/59.6	83.6/82.1	48.6/80.5	74.7/74.0	65.5/84.2	83.3/60.9	75.7/53.9	74.7/83.3	73.3
CoF-Net[10]	84.0/**83.2**	**85.3**/57.4	82.6/62.2	90.0/82.9	47.1/77.6	80.7/68.2	**73.3**/89.9	**89.3/68.7**	74.0/49.3	**84.5**/85.2	75.8
Ours	87.8/78.6	83.3/**62.9**	87.8/**63.6**	86.7/90.2	48.8/83.4	78.5/**79.4**	68.2/83.8	85.0/61.5	**76.5/67.9**	73.2/84.7	**76.6**

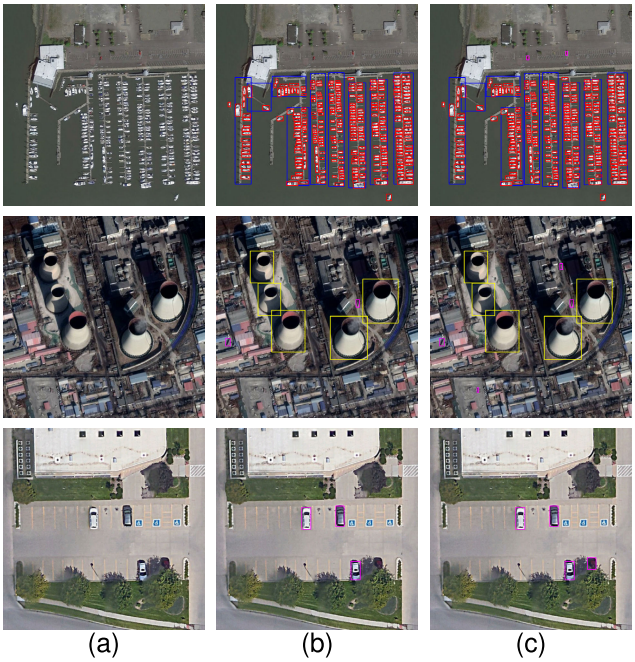


Fig. 6. Qualitative comparison results. (a) Original image. (b) Detection result of the baseline model. (c) Detection result of SFSANet.

2.5%, 11%, and 12.5%, respectively. In addition, it is worth emphasizing that SFSANet excels in accurate detection of dense small objects. For ships, the mAP of SFSANet improved by 0.6% compared to the baseline algorithm. Therefore, SFSANet has the ability to accurately detect objects with large-scale differences. In Fig. 6, we visualize the detection results of the proposed method and baseline model. It can be seen that SFSANet performs better at detecting objects in remote sensing images, which can reduce the phenomenon of missed detection. The car on the shore of the first line is not detected in the baseline, however, in the proposed algorithm, not only the ship but also the car is detected. The original drawing of the second row contains five large object tanks and four small object vehicles. The baseline detected all of the large object tanks, but only two of the small object vehicles. Our proposed method detects both large and small objects. The original image of the third row contains only four vehicles. Only three vehicles are detected at the baseline, and four vehicles are detected in our proposed method.

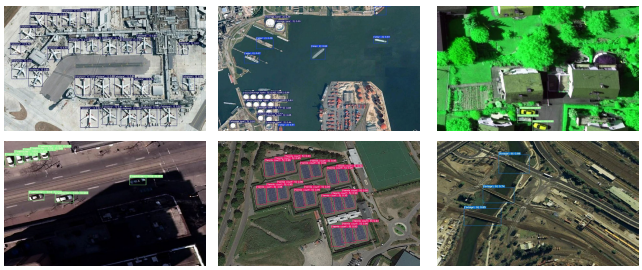


Fig. 7. Detection results of SFSANet on NWPU VHR-10 dataset.

2) Detection Results on NWPU VHR-10: Table II describes the performance comparison between the proposed method and several state-of-the-art methods on the NWPU VHR-10 test set. The results show that our method is superior to other methods, with an mAP of 92.3%. Specifically, the proposed method SFSANet has achieved remarkable results in airplane detection, with an mAP of 100%. Compared to the baseline model, we successfully improved performance by 0.4%. Additionally, in ship and vehicle detection, SFSANet excels with performance improvements of 2.9% and 0.9% in mAP, respectively, compared to the second-ranked model. Fig. 7 displays some detection results on the NWPU VHR-10 dataset, further demonstrating the outstanding performance of our method.

E. Ablation Study

To comprehensively evaluate the proposed method SFSANet, we conduct ablation experiments on the NWPU VHR-10 dataset. These ablation experiments are designed to analyze the impact of the SF, SLA, SA, and SIoU loss

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TABLE II
COMPARSION RESULTS ON NWPU VHR-10 DATASET. THE BEST RESULT IN EACH CATEGORY IS HIGHLIGHTED IN BOLD

	AP	SH	ST	BD	TC	BC	GTF	HA	BR	VE	mAP
YOLOv2[10]	83.0	84.4	81.9	84.3	85.0	53.5	62.8	78.7	85.0	70.0	76.8
Faster R-CNN[24]	94.6	82.3	65.3	95.5	81.9	89.7	92.4	72.4	57.5	77.8	80.9
YOLOv3[26]	99.6	81.3	80.3	98.3	80.6	81.82	99.5	74.3	**89.6**	87.0	87.3
YOLOv7[16]	99.6	84.8	68.5	99.1	93.3	89.9	96.6	94.3	77.4	94.1	89.2
FMSDD[35]	99.7	**89.9**	90.3	98.2	86.0	96.8	99.6	75.6	80.1	88.2	90.4
CAD-Net[11]	97.0	77.9	95.6	93.6	87.6	87.1	99.6	**100.0**	86.2	89.9	91.5
GLNet[7]	100.0	88.7	84.4	98.5	81.6	88.2	**100.0**	97.2	88.4	90.9	91.8
SCRDet[12]	100.0	89.4	**97.2**	97.0	83.2	87.5	99.2	99.4	74.5	90.1	91.8
Ours	**100.0**	87.7	74.4	**99.4**	**95.9**	**93.9**	98.9	97.1	80.9	**95.0**	**92.3**

TABLE III
ABLATION EXPERIMENTS ON NWPU VHR-10 DATASET

Baseline	SF	SLA	SA	SIoU	P/(%)	R/(%)	mAP/(%)	Params(M)	GFLOPs	FPS(s)
✓					90.4	83.0	89.2	6.0	13.3	65
✓	✓				91.3	86.8	90.7	7.1	18.8	59
✓		✓			90.8	85.4	91.2	6.7	16.4	57
✓			✓		92.6	87.6	91.8	7.4	19.2	53
✓	✓	✓			91.6	87.5	91.5	7.9	20.1	49
✓	✓	✓	✓		92.3	88.9	92.0	9.1	24.3	46
✓	✓	✓	✓	✓	93.2	89.0	92.3	9.1	24.3	46

function on the final experimental results. The results are shown in Table III. The specific analysis of the ablation experiment results is as follows.

1) Effect of SF: We add SF module to the baseline model, which improved the mAP of the NWPU VHR-10 dataset from 89.2% to 90.7%, an increase of 1.5%. The experiment proves that the introduction of SF module is effective, which is helpful to improve the detection accuracy of the model and enrich the semantic information. This module uses matrix multiplication operation to calculate the correlation between pixels in the same feature map and the correlation between pixels in different feature maps to improve the ability to classify and locate.

2) Effect of SLA: The SLA is added to the baseline to increase sensitivity to the object. It can be seen that the mAP increased from 89.2% to 91.2%. The experimental results show that to deal with the problem that the objects are easily submerged in the background, the parallel structure design is adopted to build SLA module, which can effectively make the model focus more on the object information and effectively suppress unnecessary feature information.

3) Effect of SA: Building upon the results of the fourth set of experiments, the addition of the SA module increased the mAP from 89.2% to 91.8%, resulting in an improvement of 2.6%. The effectiveness of the SA module is proven, because it can not only retain local details but also capture more global information, which can improve the utilization of feature information.

4) Effect of SIoU: We adopt the SIoU loss function as the localization loss function and finally increase the mAP from 92.0% to 92.3%, an increase of 0.3%. It is worth noting that during the training, the SIoU loss function not only improves the model performance but also speeds up the model convergence speed.

V. CONCLUSION

In this article, we propose the remote sensing image object detection method based on SF and SA. First, the SF module is proposed to weaken the semantic differences between different scale feature maps and enrich the semantic information. Second, the SLA module is constructed to enhance the attention of the detection model to the object and effectively suppress the interference caused by background and noise. Third, the SA module is designed to capture multiscale feature information and improve the effective utilization rate of object feature information. Finally, we optimize the localization loss function with the SIoU loss function to accelerate the convergence speed of the model. Experimental results demonstrate that on the DIOR and NWPU VHR-10 remote sensing datasets, our method achieves mAP scores of 76.6% and 92.3%, respectively, outperforming the comparison algorithms in object detection performance. However, we use horizontal bounding boxes to locate detected objects, but this method is not perfect. In the future, we will explore a more efficient and accurate detection approach to better suit remote sensing image object detection.

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